

Nitant Rai

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PROFESSIONAL SUMMARY

Ph.D. Candidate in Forestry at Virginia Tech, concurrently completing an M.Eng. in Computer Science. Develops Python and R workflows for UAV imagery, LiDAR, computer vision, and forest monitoring, connecting field data collection with geospatial analysis and model validation.

TECHNICAL SKILLS

Programming & Libraries: Python, R, SQL, PyTorch, scikit-learn, NumPy, Pandas

Geospatial Tools: ArcGIS Pro, Google Earth Engine, GeoPandas, GDAL, rasterio

Methods: LiDAR processing, UAV imagery, computer vision, statistical modeling, forest inventory

Development & Validation: Git/GitHub, reproducible workflows, quality checks, uncertainty analysis

EXPERIENCE

Graduate Research Assistant

Jan. 2023 – Present

Virginia Tech

- Build reproducible Python and R workflows integrating raster, point cloud, and field inventory data for forest structure assessment and regeneration monitoring.
- Apply machine learning and computer vision to forest measurement tasks; evaluate outputs using statistical validation, quality checks, and uncertainty analysis.

Drone Park Intern

May – Aug. 2025

Virginia Tech

- Planned UAV flights, maintained equipment, and supported technical training and demonstrations for research and educational applications.

Graduate Research Assistant

2021 – 2022

Mississippi State University

- Processed airborne and ICESat-2 LiDAR for canopy height estimation, assessed measurement uncertainty, and delivered backpack LiDAR workshops.

Teaching experience, Virginia Tech: GIS analysis, forest inventory, and field measurement instruction for students with varied technical backgrounds.

SELECTED PROJECT

UAV Seedling Detection and Monitoring Pipeline | *Python, YOLO* | [GitHub](#)

- Developed batch image inference and detection-count reporting for UAV-based forest regeneration monitoring, with CSV summaries and visual outputs.

EDUCATION

Virginia Tech

2023 – Present

Ph.D. Candidate in Forestry; M.Eng. in Computer Science, in progress

Mississippi State University

2021 – 2022

M.S. in Forestry; Minor in Geoscience

Tribhuvan University

2014 – 2018

B.S. in Forestry

CERTIFICATION

FAA Part 107 UAS Remote Pilot certification.